

# Curriculum vitae

## Karthikeyan Govindan

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### Experience

- 1. Research Officer** - Wellcome Trust Research Laboratory, Christian Medical College | Vellore, Tamil Nadu, India | 05/2022 – Present
- 2. Project Assistant** - King Institute of Preventive Medicine and Research | Chennai, Tamil Nadu, India  
03/2021 – 02/2022

### Research specialization

- Wastewater Genomic Surveillance
- Gut microbiome
- Pathogen sequencing
- Functional Genomics
- Bioinformatics

### Publications

1. Exploring Gut Mucosal Microbiota to Distinguish Crohn's Disease from Intestinal Tuberculosis. *Inflammatory bowel diseases*. 2026
2. Dynamics of tissue and fecal microbiota in active Crohn's disease and their ability to predict disease state. *World Journal of Gastrointestinal Pathophysiology*. 2025

3. AmpliconTyper-a tool for analysing ONT multiplex PCR data from environmental and other complex samples. *Microbial Genomics*. 2025
4. EnviroAmpDesigner—a tool for designing high specificity multiplex PCR primer panels for detecting and subtyping a target organism in environmental surveillance samples. *bioRxiv* (2025).
5. Unveiling the Global Surge of Mpox (Monkeypox): A comprehensive review of current evidence. *The Microbe*. 2024.
6. qPCR based multipathogen detection for SARS-CoV-2, CrAssphage and Hepatitis E virus from wastewater samples. *Protocols.io*. 2024
7. Distribution and Functional Analyses of Mutations in Spike Protein and Phylogenetic Diversity of SARS-CoV-2 variants emerged during 2021 in India. *Journal of Global Infectious Diseases*. 2023.
8. Epitope identification and design of a potent multi-epitope vaccine construct against SARS-CoV-2 including emerging variants. *Journal of Global Infectious Diseases*. 2022.
9. Serum levels of matrix metalloproteinases as prognostic markers for severe dengue with plasma leakage. *Experimental and Molecular Pathology*. 2022.
10. The utility of inflammatory and endothelial factors in the prognosis of severe dengue. *Immunobiology*. 2022.
11. Phylodynamic pattern of genetic clusters, paradigm shift on spatio-temporal distribution of clades, and impact of spike glycoprotein mutations of SARS-CoV-2 isolates from India. *Journal of Global Infectious Diseases*. 2021.

## Awards & Scholarships

1. **Bursary Awardee**, Wellcome Connecting Science — *Understanding Life: Using Large-Scale Biodiversity Reference Genomes* (2025)
2. **Bursary Awardee**, Oxford Nanopore Technologies (2025)

## Current Research Projects

1. Novel tools for measuring typhoid vaccine impact (R01) [ Funding - NIH ].
2. Optimisation of an integrated protocol for the molecular detection and sequencing of priority viral and bacterial pathogens in environmental samples [ Funding - BMGF ]

3. Standardization of methods and models for multipathogen detection from wastewater in Vellore [ Funding - BMGF ]
4. Elimination of Malnutrition through Action-Research on Moderately Acute Malnourished Children (CSR Grant)A - Sun Pharma Ltd.
5. Vellore Typhoid ES for TCV impact assessment [ Funding - BMGF ]
6. Establishing sewage surveillance and laboratory testing methods for the environmental surveillance of COVID-19 [ Funding - BMGF ]
7. Long term stability of gut microbiota in patients with Crohn's disease and correlation with biomarkers of inflammation and gut permeability [ Funding - DST India ]
8. Genetic attributes and virulence of Salmonella Typhi impacting its intracellular survival within Acanthamoeba spp. - Grand Challenges Explorations grant [ Funding - BMGF ]
9. Estimation of Rotavirus Evolutionary Divergence of Time Using Bayesian Evolutionary Analysis (BEAST).

## Completed Research Projects

1. Evaluation and characterization of crucial biomarker profiles for stratifying Dengue disease severity [ Funding - ICMR ]
2. Diagnosis of SARS-CoV-2 and detection of other viral respiratory pathogens in SARI patients.
3. Spatiotemporal analysis and deep mutational scanning of Indian SARS-CoV-2 genomes.
4. Establishment of human primary Gingival fibroblast and primary Dermal fibroblast cell lines.

## Education

King Institute Of Preventive Medicine And Research | Chennai, Tamil Nadu, India

**Master of Science (MSc)** - Molecular Virology | 07/2021 | Percentage: 75.5

Vels Institute Of Science Technology And Advanced Science | Chennai, Tamil Nadu, India

**Bachelor of Science (BSc)** - Microbiology | 07/2018 | CGPA: 7.88

## Presentations

1. Chromosome scale reference genome model - *Acanthamoeba polyphaga* (ATCC-30461) – T4 genotype, *Understanding Life: Using Large-Scale Biodiversity Reference Genomes*, Wellcome Connecting Science (October 2025)
2. Functional Characterization and Reference Genome Assembly of *Acanthamoeba polyphaga* (ATCC-30461) – T4 genotype, BioDiversity Genomics Conference 2024 (BG24), Tree of Life Programme, Wellcome Sanger institute. October 2024.
3. Composition and Temporal Dynamics of Microbial Communities in Drinking Water from Peri-Urban Areas of Vellore. Winter Symposium, Christian Medical College and Hospital, Vellore. (February 2024).

## Skills

### Technical skills

- Sequencing - MiSeq (Illumina), MinION (ONT)
- PCR- Conventional PCR , Real time PCR.
- ELISA.
- Nucleic Acid extraction.
- Gel electrophoresis.
- Qubit Fluorometer/spectrophotometry.
- Tapestation (Fragment analyzer).
- Fluorescence Microscope.
- Cell Culture / Virus Isolation

### Bioinformatics Skills

- NGS QC: FastQC, MultiQC, FastP, Trimmomatic, Quast.
- Genome Assembly: SPAdes, Flye.

- Genome Binning: MetaBAT2, MetaWrap, MetaCoAg, Maxbin.
- Genome Annotation: Prokka, Augustus, BRAKER3, Artemis.
- Functional Annotation: KEGG, KO, GO, eggNOG, InterPros, AlphaFold.
- Transcriptomics: Hisat2, StringTie, DESeq2, Cufflinks.
- Phylogenetics: RAxML, FastTree, BEAST, MEGA.
- Metagenomics: Qiime2, DADA2, metaSPAdes. Megahit.
- Classifier: 16s and 18s Silva and PR2 classifiers.
- Workflow management: Nextflow, Snakemake and Bash.
- Dashboards & Real-time Visualization: Shiny | GitHub Pages | ARTIC RAMPART

## Computing Skills

- High Performance Computing(HPC) - Google Compute Engine | AWS cloud computing
- Linux – CPU | GPU parallelization
- Coding language - Bash | R
- Google Colab Notebook (Python and R/bash)
- GitHub - version control Codes | Documents | Data
- Containerization - Docker | Pipelines | Scripting | Automation
- Document management – R Markdown | Quarto | GitBook
- Web servers: Chan Zuckerberg ID | Galaxy servers | BV-BRC
- Web site development : Github pages.

## Certificates

- Certificate of Appreciation for Organizing the *Galaxy Training Academy 2025*.
- Analyzing and Visualizing NextGen Reads with Magic-BLAST (NIH- NCBI)
- Bioinformatics for Biologists - An Introduction to Linux, Bash Scripting and R ( Wellcome connecting science ).

- Eukaryotic Genome Assembly - How to use BlobToolKit for quality assessment ( Wellcome connecting science ).
- EMBL-EBI- Training online Courses.
- A Practical guide for SARS CoV-2 Whole Genome Sequencing ( Wellcome connecting science ).
- From swab to server: Testing , Sequencing and Sharing During a Pandemic ( Wellcome connecting science ).
- Bacterial Whole Genome Sequencing and Advanced Genomic Data Mining.
- Nanopore Community Meeting (NCM) 2022.
- Second virtual training on KFD virus Biosafety, molecular and serological methods (NIV-pune).
- Trained on vector virus relationship, Epizootiology of vector borne viral diseases, Molecular Entomology, Mosquito species identification, collection, processing , preservation and dissection of mosquito. (ICMR-VCRC).

## Contact info

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6. **Google Scholar** ID - <https://scholar.google.com/citations?hl=en&user=is4cNvYAAAAJ>